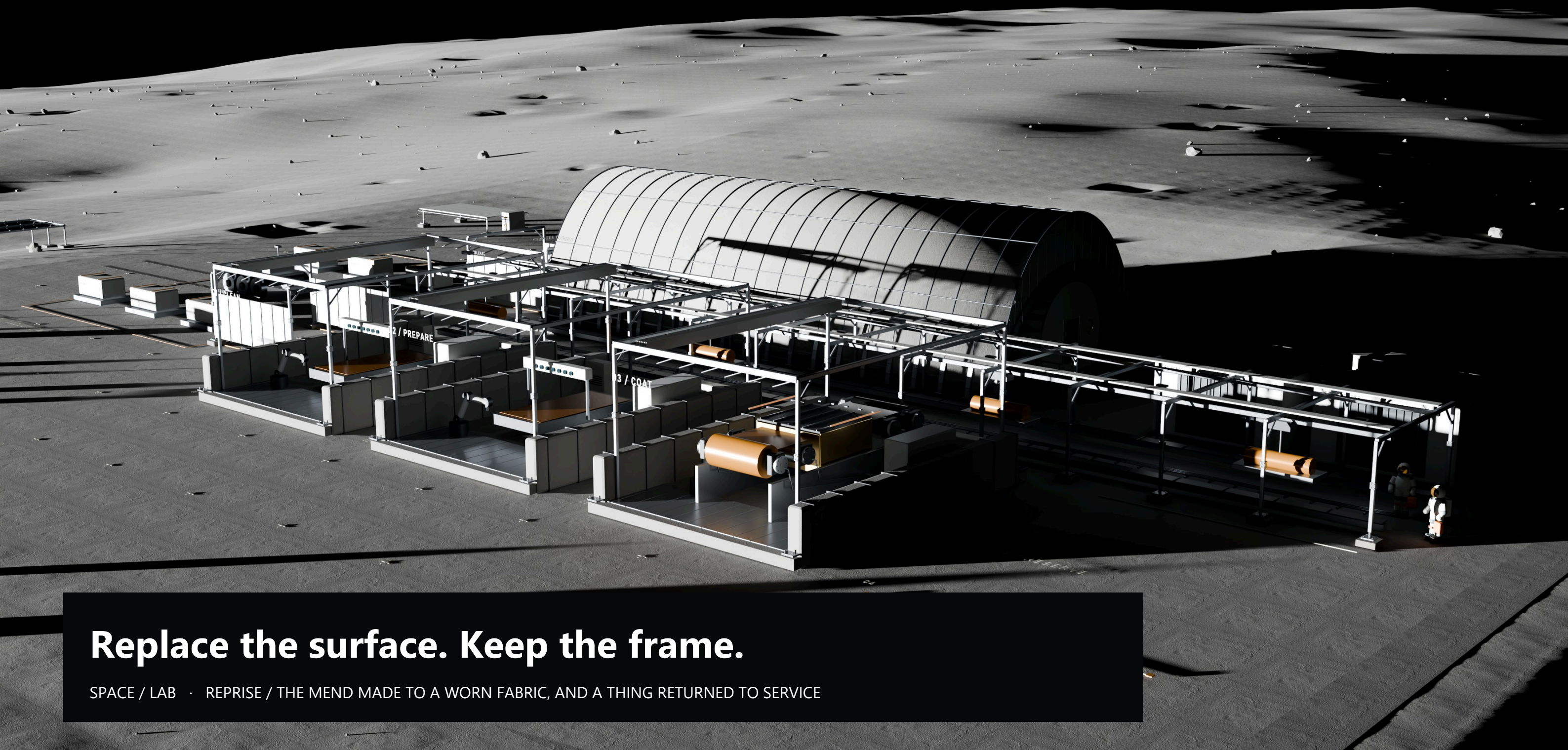


# REPRISE

MAKE SOLAR REFLECTORS. BUILD A CULTURE OF REPAIR.



**Replace the surface. Keep the frame.**

SPACE / LAB · REPRISE / THE MEND MADE TO A WORN FABRIC, AND A THING RETURNED TO SERVICE

## 01 / JURY SYNTHESIS: PRODUCT, PEOPLE AND FIRST TEST

# Make, repair and inhabit the Moon.

Reprise is a lunar workshop for making, repairing and reusing solar reflectors. Its first product redirects sunlight with two replaceable film strips on a reusable 50 m<sup>2</sup> frame.

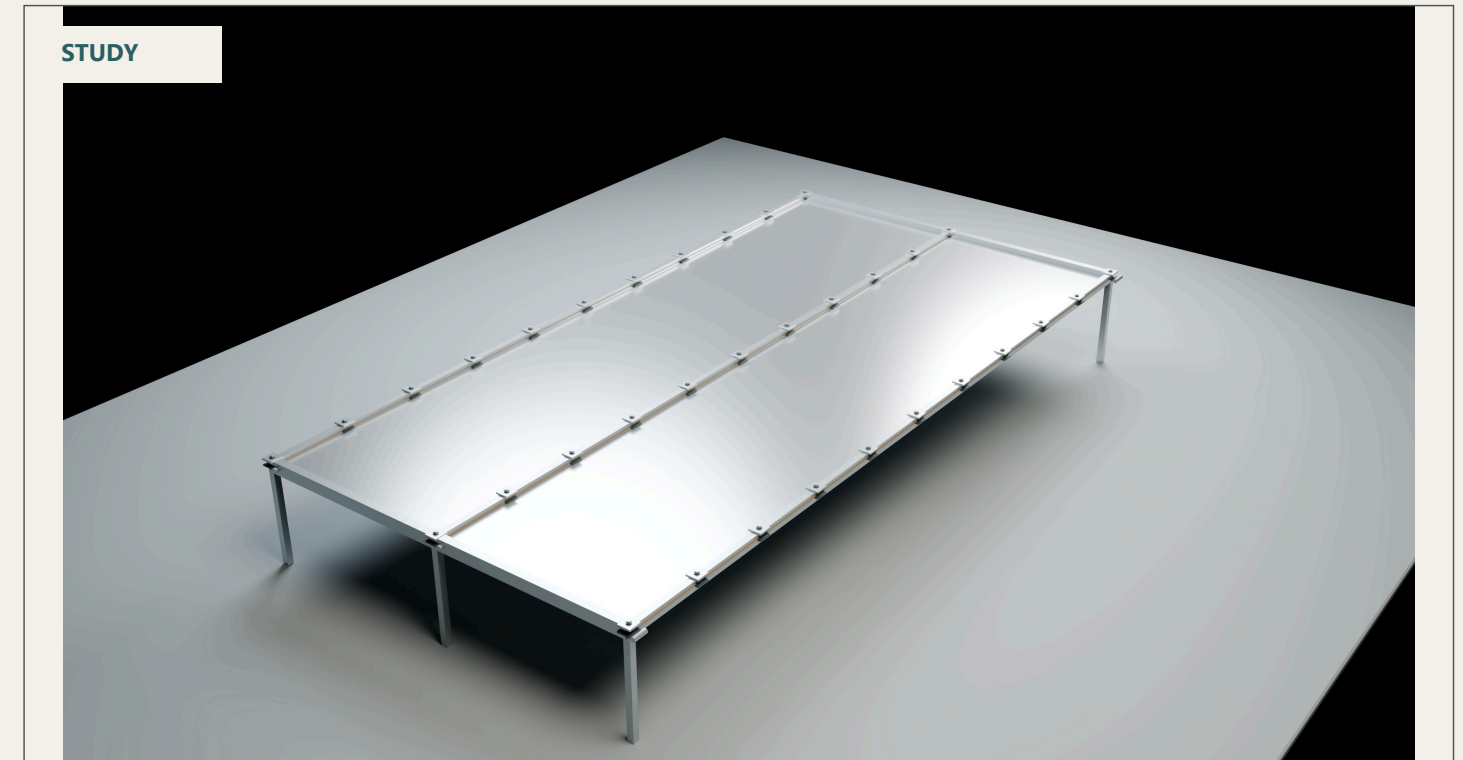
Three robotic courts meet a covered transfer street. Beside them, a shielded workplace brings human diagnosis, skilled repair and shared decisions to the machines: a productive anchor for a permanent lunar community.



## THE PROJECT ONLINE

[marc-dvci.github.io/reprise](https://marc-dvci.github.io/reprise)

Film, both books, the A0 board and the ten illustrations



## A PLACE TO MAKE THINGS LAST.

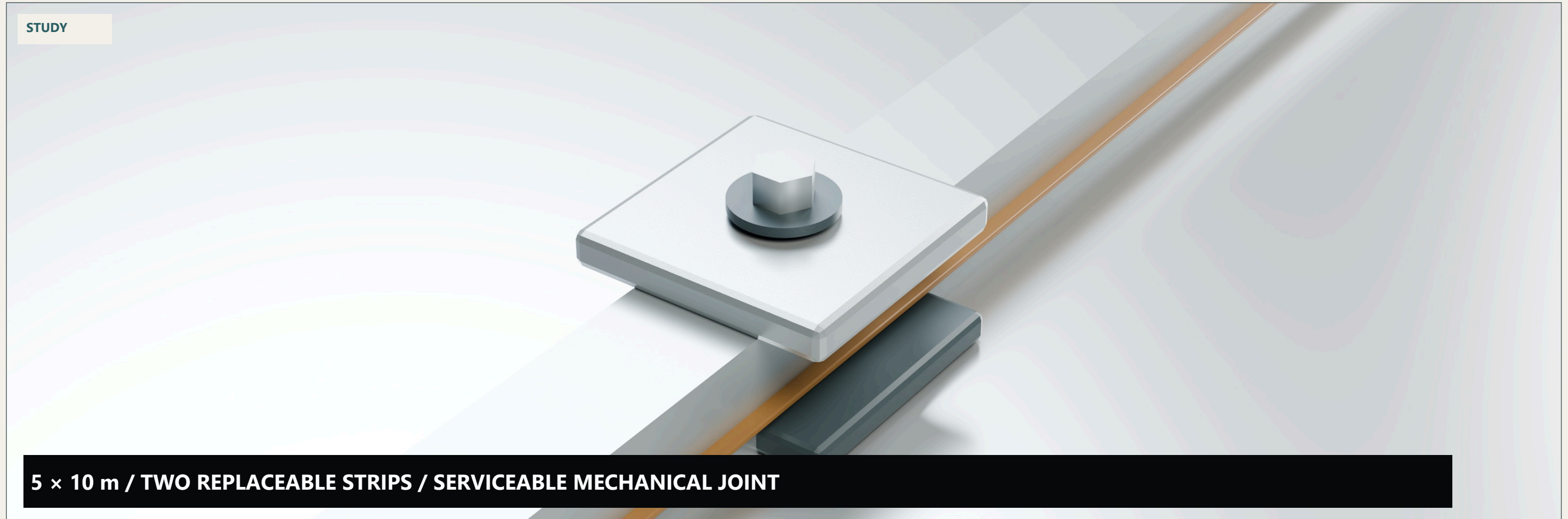
Robots renew surfaces in vacuum. Specialists rebuild removable machine parts at repair benches, inspect samples and approve reuse. The Earth Lab develops coating quality and repeated renewal; the architecture turns maintenance into a shared daily practice.

Architectural proposal / Earth demonstrator: p. 9; opportunities and scale: p. 13; resources: p. 14. Current crew sizing covers visiting missions.

## 02 / PRODUCT, OPTICAL USE AND REPAIR

# Replace worn film and reuse the reflector frame.

STUDY



5 × 10 m / TWO REPLACEABLE STRIPS / SERVICEABLE MECHANICAL JOINT

## HOW A PASSIVE REFLECTOR WORKS

SUNLIGHT

TOWARD A  
RECEIVER

Sunlight redirected to a receiver; the Earth Lab develops pointing and surface accuracy.

A 50 m<sup>2</sup> passive reflector redirects sunlight. Mechanical clamps grip reinforced film edges; the precision frame is reused.

25 µm polyimide  
100 nm target aluminium thickness  
50 N/m pretension  
≈ 15 mm calculated gravity sag  
80 mm tensioner travel

1.79 kg coated skin, excluding frame and reinforced edges. Joint detail study. Recoat sound film; replace worn strips; repair and retain the frame.

## 03 / A PRODUCTIVE FOOTHOLD FOR A LUNAR COMMUNITY

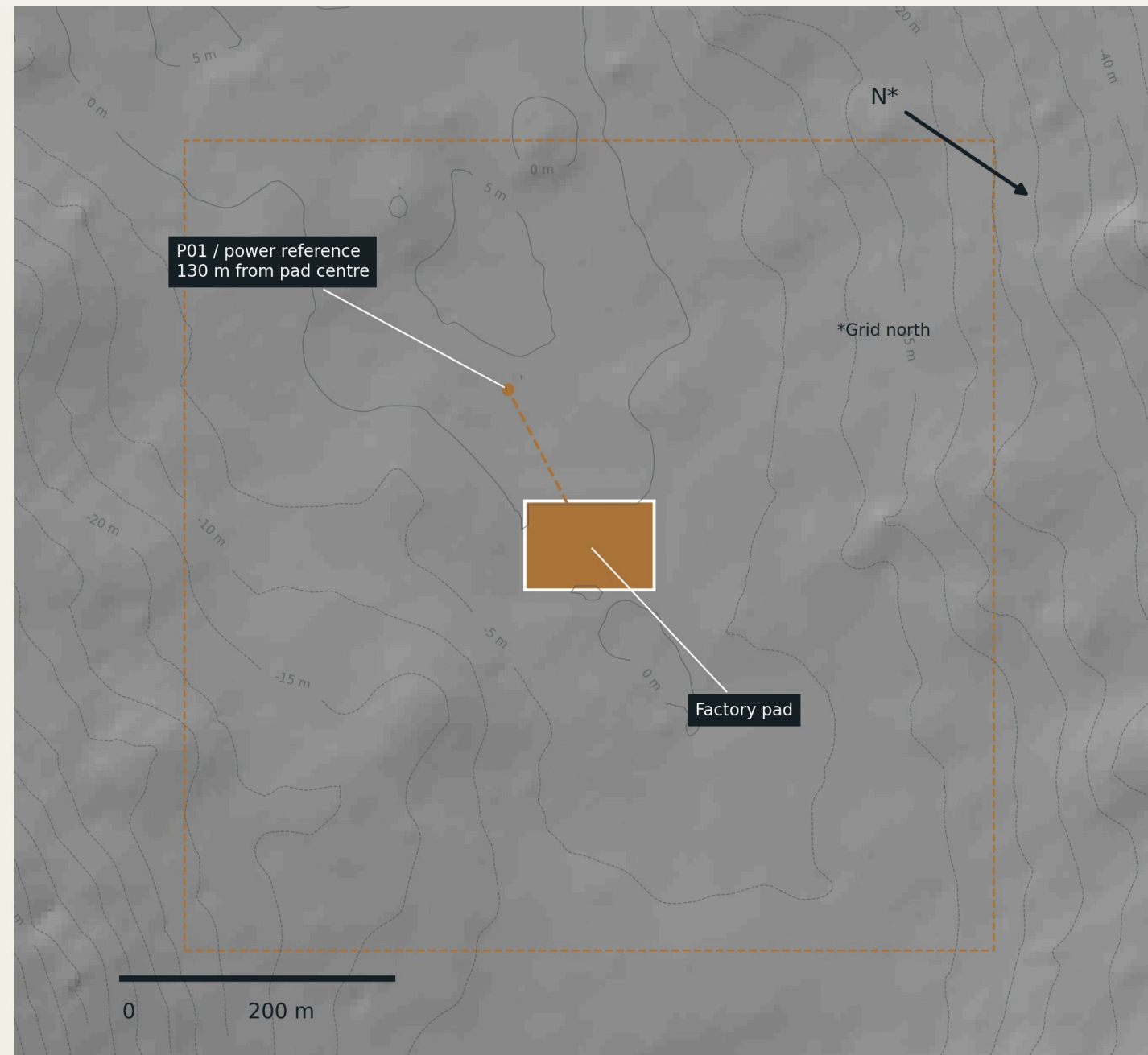
## Connecting Ridge shapes the architecture.



GLOBAL LUNAR CONTEXT  
NASA LRO near-side view  
Site: 89.458° S, 222.879° E

First crew module: 2 specialists · six 8 h visits/year  
Ambition: recurring repair work within a permanent settlement

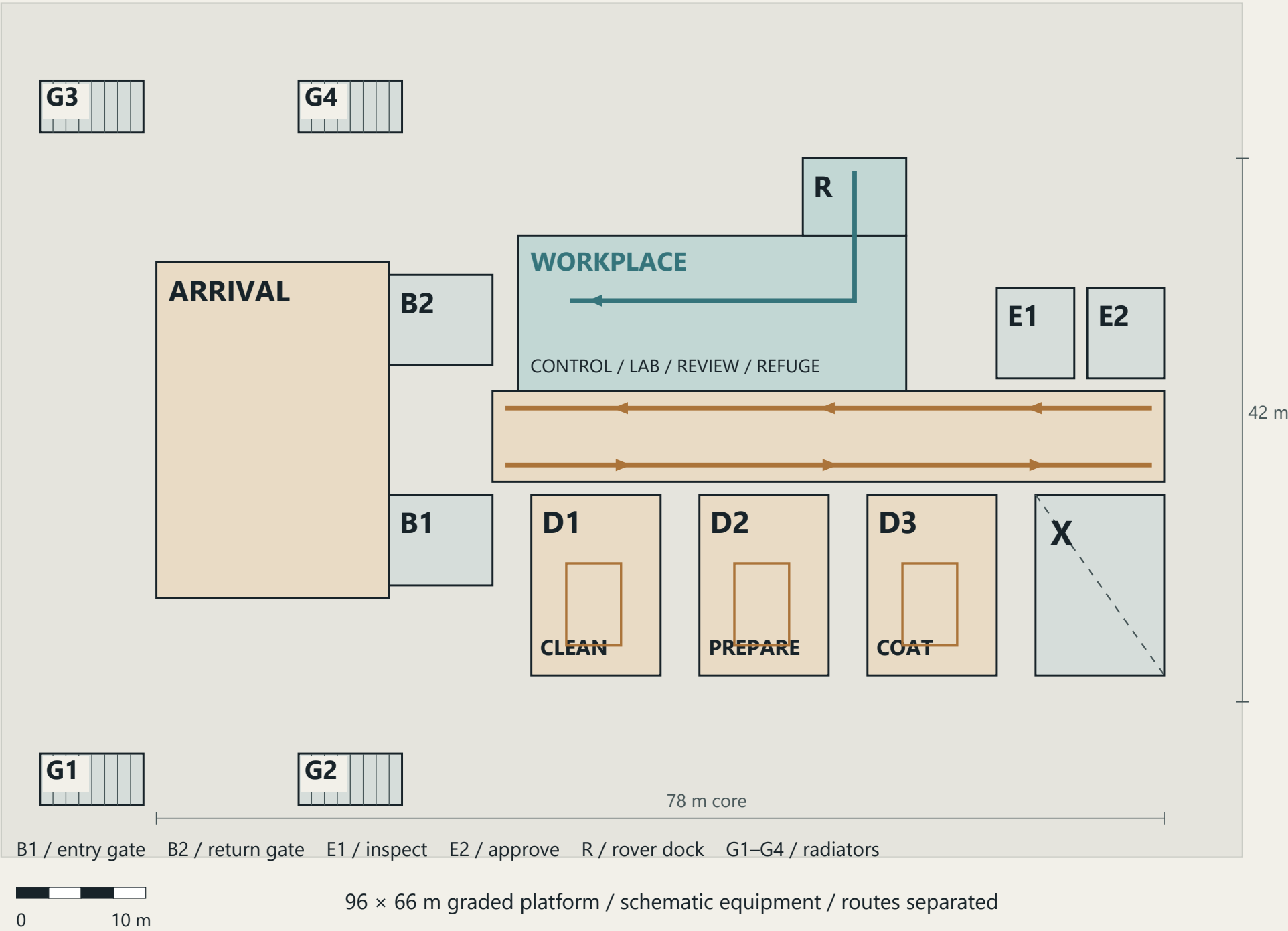
NASA PGDA Site 01 / P01 is the solar-power reference. Its sampled 39 h darkness interval informs the energy-storage study.



5 m terrain grid · 96 × 66 m graded pad  
P01 power reference: 130 m from pad centre

04 / PROGRAMME, SPATIAL ORDER AND CIRCULATION

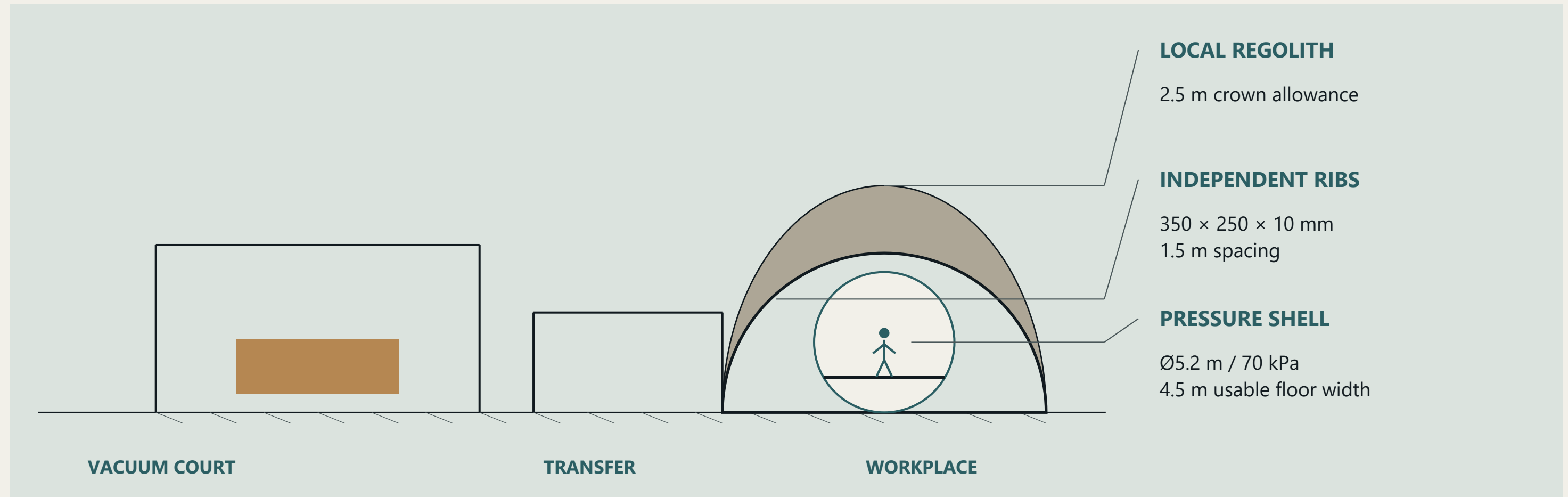
# Separate routes for materials and crew.



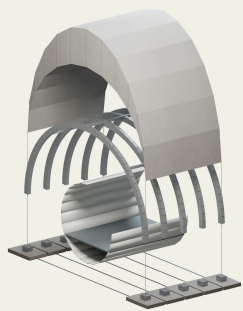
Amber: materials circulate through the vacuum courts. Teal: people enter the shielded workshop from the rover. X reserves space for shared services.

## 05 / CONSTRUCTION, MATERIAL AND LOAD PATH

# An independent frame supports the regolith shield.



## STUDY



## 01 / BEARINGS + TIES

Set out supports and buried 800 mm<sup>2</sup> ties. Assemble braced ribs at 1.5 m centres.

## 02 / FRAME + LOCAL FILL

350 × 250 × 10 mm aluminium ribs carry retaining panels and compartmented regolith.

## 03 / PRESSURE MODULE

Commission the independent Ø5.2 m shell. No regolith load is assigned to it.

Crew-workplace structure / Exploded 6 m bay. Preliminary rib stress ≈49 MPa; deflection ≈24 mm. Separate layers make components accessible for replacement.

06 / HUMAN EXPERIENCE IN THE PRESSURISED WORKPLACE

# A workshop with a human rhythm.



## HANDS, LIGHT AND SHARED TIME

From the rover threshold, a lit spine leads to repair benches and a shared sample table. Specialists diagnose wear, rebuild removable mechanisms and authorise reuse. Warm task light and acoustic panels give way to a quiet retreat. A live exterior view reconnects this sheltered interior to the polar landscape.



CONTROL / ORIENT

LAB / FOCUS

WORKSHOP / REPAIR

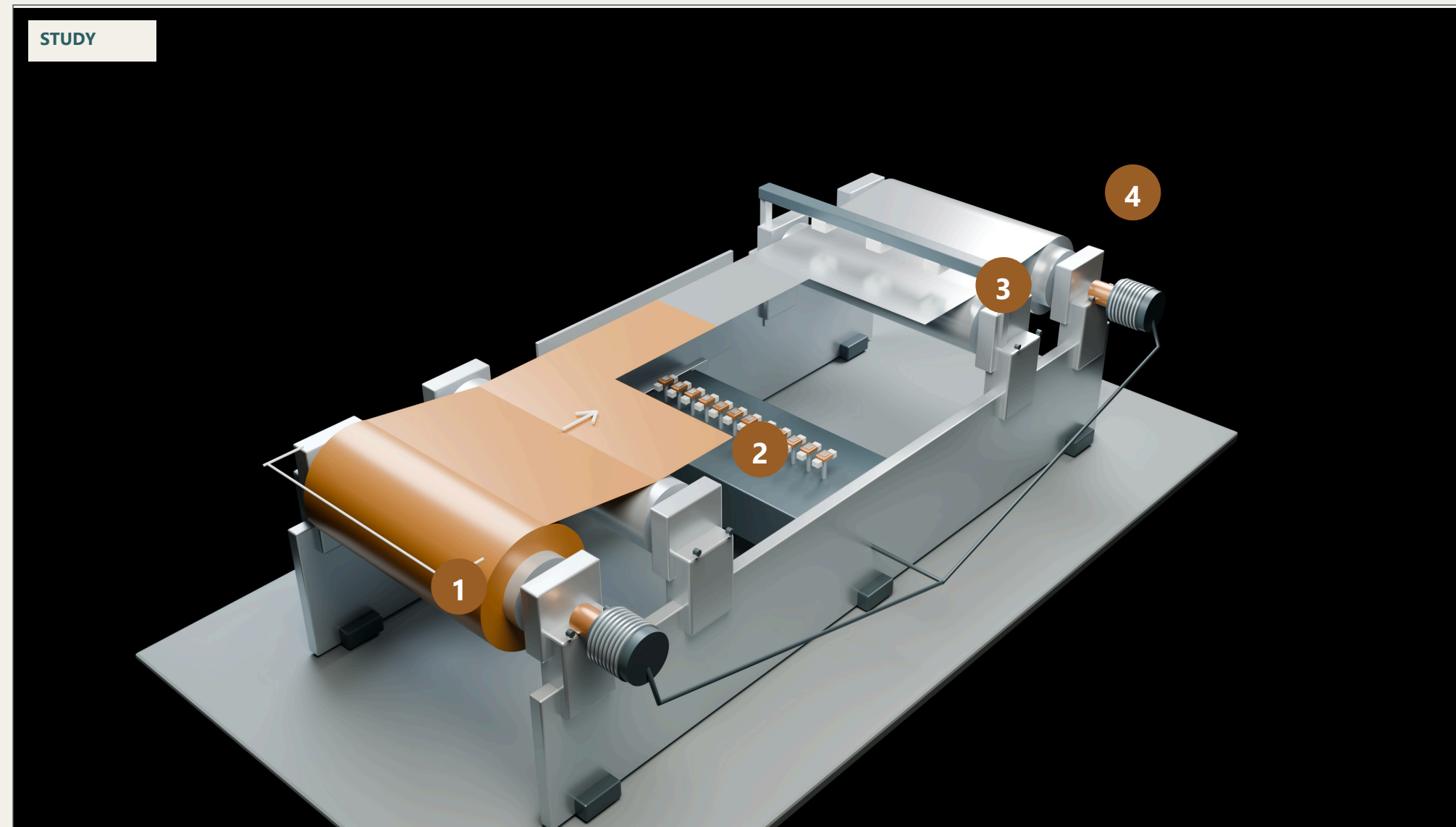
SHARED TABLE / MEET

QUIET ROOM / WITHDRAW

30 m workplace / 135 m<sup>2</sup> gross floor / 2 visiting specialists. Permanent settlement presence is the ambition; the current life-support study covers visits.

## 07 / COATING AND QUALITY CONTROL

# From incoming film to an inspected roll.



## 2.5 m FILM WIDTH

01 / Unwind polyimide

02 / Deposit aluminium

03 / Inspect the surface

04 / Rewind and label the roll

100 nm aluminium target / twelve evaporation sources.  
900 m<sup>2</sup>/h nominal film travel capacity. A 3,000 m<sup>2</sup>/year scenario equals  
60 reflector skins of 50 m<sup>2</sup>.

Vacuum enables thin-film deposition near the point of use. Fresh coating  
and local renewal offer quality opportunities; the Earth Lab measures  
reflectance and usable output.

Cutaway: hood and part of film omitted; neutral inspection lighting. Joining film to the reusable frame is a separate operation, shown on page 3.

## 08 / EARTH DEMONSTRATOR: COATING, REPAIR AND REPEATED REUSE

# Develop the art of renewing a surface.

## A / EARTH-COATED CONTROL

Coat + inspect → package → transport / storage exposure → unpack → install.

## B / COAT AFTER DELIVERY

Package bare film → same exposure → clean + coat → inspect → install. Include imported Al and process losses.

## C / RENEW RETAINED FILM

Recover → assess the substrate → clean → recoat → inspect → reuse. Replace strips at the end of their useful life.



**Reuse the frame. Renew sound film. Repair the machines that make it.**

## COMMON BASIS

Same polymer batch, frame, optical duty and availability. Optimise packaging for both new-film routes. Record accepted area and replacement time.

## MASS AND SERVICE

Measure package, feedstock, rejects, spares, full-cycle energy and labour per accepted m<sup>2</sup>. Test renewal life separately; frame reuse is common.

## SCREENING TARGETS

3 independent runs; 100 nm ±10% thickness; solar reflectance ≥0.85 and within 0.02 of control. Receiver intercept ≥95% of control at the same geometry.

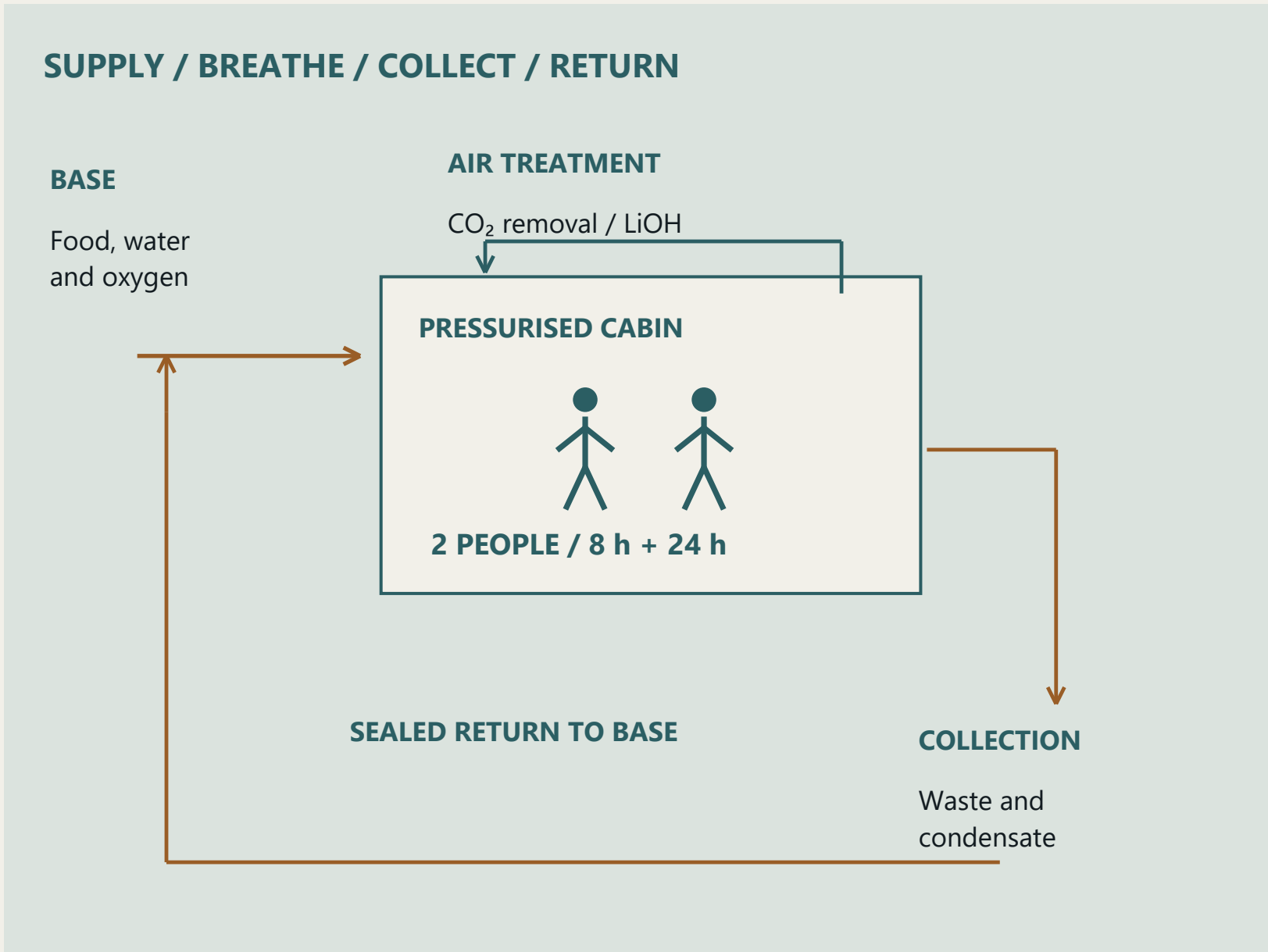
## CLEAN PRODUCTION

Capture ≥99% of material missing the film. Measure source utilisation and deposition around the equipment across repeated cycles.

Proposed Earth Lab targets / Compare with Earth-coated reference film at equal optical service. Physical trials develop the coating and renewal process.

09 / FIRST INHABITED STEP: TWO PEOPLE, AN 8-HOUR SHIFT

# Supporting people at work.



## ATMOSPHERE

70 kPa / 30% O<sub>2</sub>

Design atmosphere for the visiting crew module.

## OXYGEN / CO<sub>2</sub>

4 kg O<sub>2</sub> / 2 × 7 kg LiOH

Cabin gas is separate; each cartridge covers all 32 h.

## WATER / FOOD

24 L / 3 kg dry food

Stored consumables; drinking and food preparation need ≈9.4 L.

## WASTE CAPACITY

32 L liquid capacity

12 L solids capacity

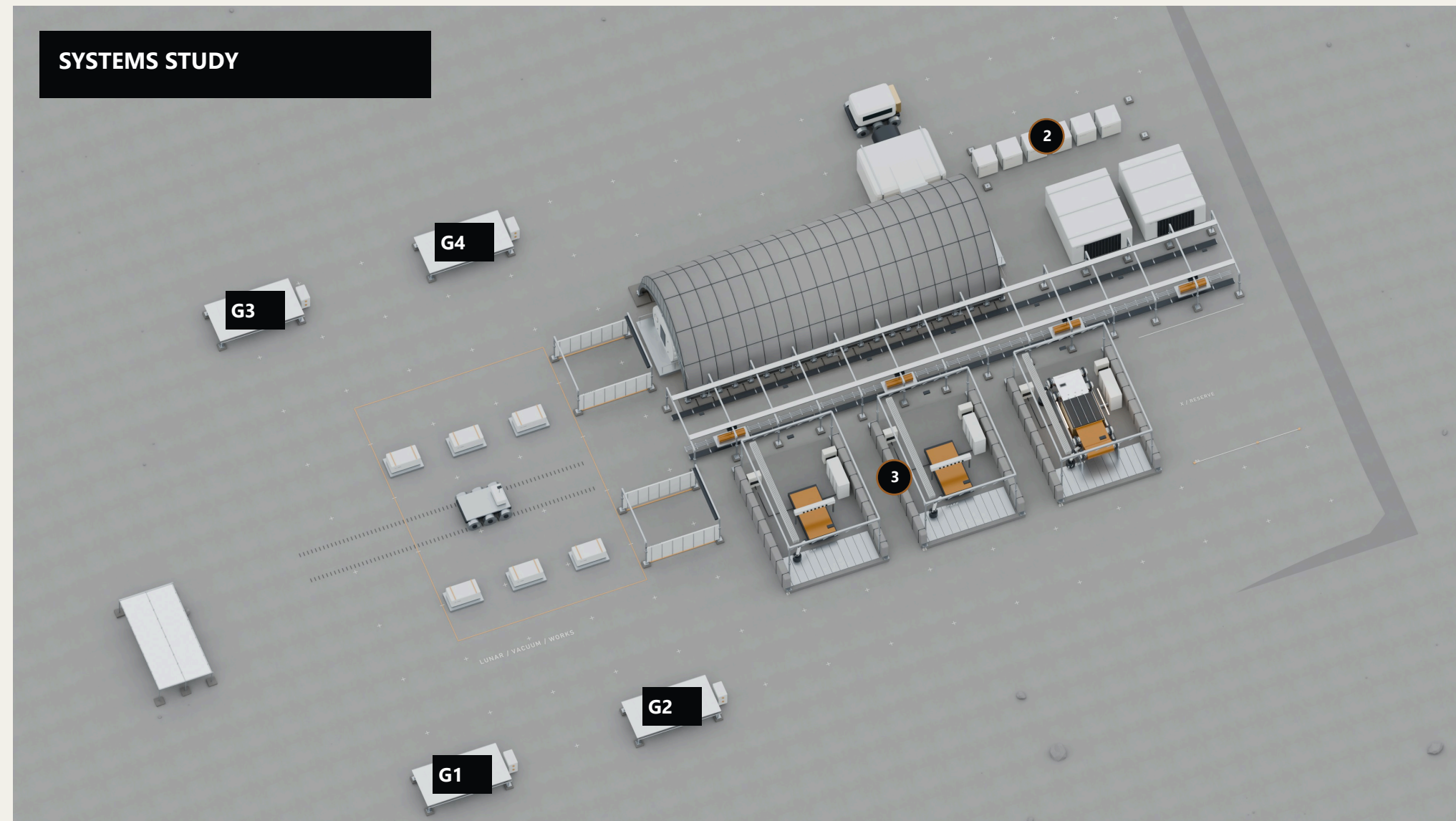
Selected container capacity, not a production quantity.

Air circulates through CO<sub>2</sub> removal. Supplies arrive from the settlement; waste and condensate return there.

Visit-sized stocks: 8 h work + 24 h reserve. Permanent residence would add continuous life support, sleeping quarters and regular resupply with the settlement.

## 10 / SOLAR COLLECTION, STORED ENERGY AND COOLING

# Energy for the courts and the crew.

**1 / COLLECT**

80 kW rated solar capacity → 60 kW net design supply at solar reference P01.

**2 / STORE**

6 × 50 kWh cabinets = 300 kWh. Usable end-of-life energy: 216 kWh.

**3 / USE**

29 kW production + 6 kW crew + 14 kW heat-pump power = 49 kW.

**G1–G4 / REJECT**

4 × 32 m<sup>2</sup> = 128 m<sup>2</sup>. Design heat rejection: 51.2 kW at 400 W/m<sup>2</sup>.

**NORMAL OPERATION**

35 kW heat + 14 kW compressor = 49 kW.

Darkness: 5 kW × 39 h = 195 kWh.

Design targets / Extended darkness: stop coating, return crew, preserve equipment. The 4 kW unoccupied mode requires thermal-vacuum testing; see photo 10.

**ONE BATTERY ISOLATED**

Five cabinets: 180 kWh usable.

Unoccupied 4 kW target: 45 h.

**ONE RADIATOR ISOLATED**

20 kW production + 6 kW crew;

36.4 kW rejection / 38.4 kW available.

## 11 / HUMAN SKILLS, MATERIAL CYCLES AND SPATIAL THRESHOLDS

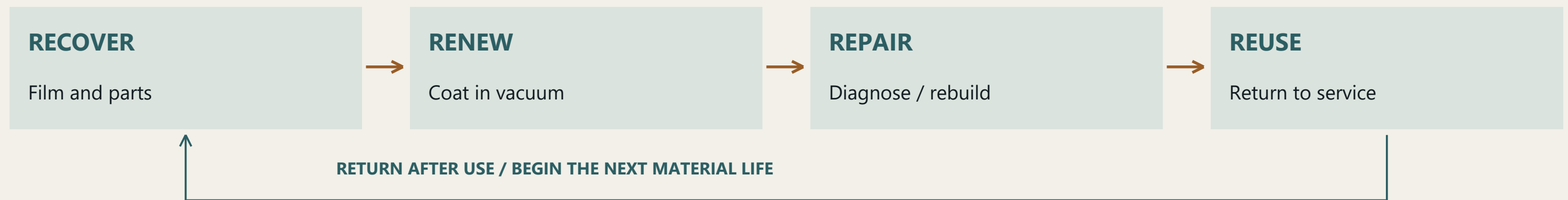
# Repair becomes a way of inhabiting.



## HUMAN SKILL / SHARED SPACE

People diagnose unfamiliar wear, adjust clamps and rebuild removable cartridges. Tools, reach and lighting make skilled work part of the spatial programme.

The sample table is both a place of precision and a meeting place. Work is reviewed together; repair knowledge passes between visiting teams.



Architectural intent / The vacuum courts handle surfaces; enclosed transfer brings removable parts to human repair benches in the pressurised workshop.

## 12 / PHASED PLANS, USEFUL SURFACES AND SHARED LUNAR INDUSTRY

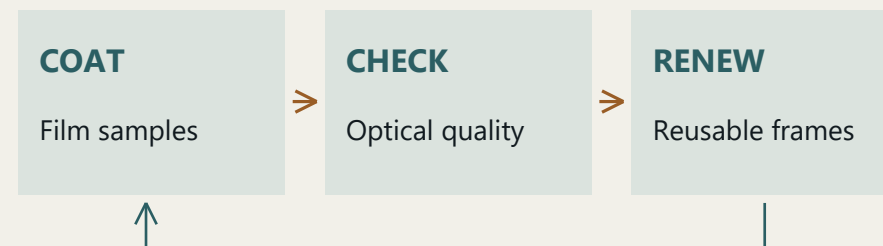
# One site. Three steps towards shared use.

From film samples to a shared lunar workplace: each step adds a useful service, keeps materials in circulation and passes repair knowledge to the next.

## 01 / EARTH LAB



### LEARN THROUGH COATING AND REUSE



### A COMPLETE MATERIAL-RENEWAL CYCLE

Develop coating quality and practise removing, replacing and re-tensioning film.

Earth-based team / no lunar modules  
Crew on Moon: 0 / Cargo landings: 0

## 30,000 m<sup>2</sup>

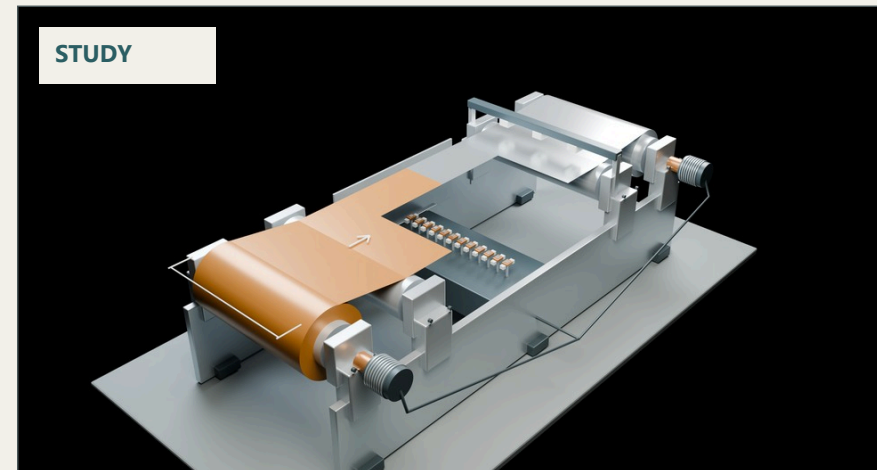
3,000 m<sup>2</sup>/year × 10 years  
Planning service scenario

**17.3 t robotics + 51.9 t crew spaces + 10.5 t solar / utilities / rover allowance = 79.7 t**

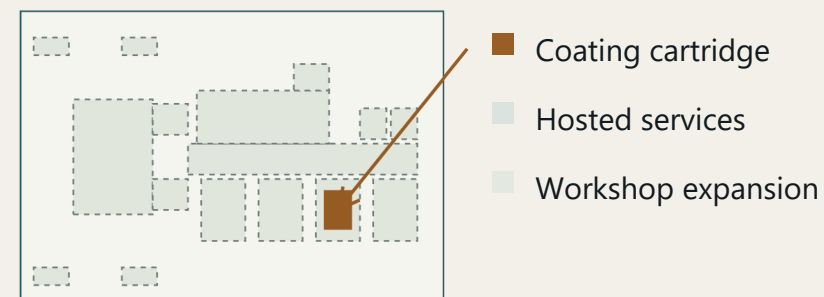
\*79.7 t ÷ 12–15 t payload = 6–7 loads by mass only (NASA cargo concept, p. 15); pilot, packing and offloading are additional. No price threshold is asserted.

MARC DONOVICI · FRANCE / SPACE / LAB / CONNECTING RIDGE

## 02 / LUNAR PILOT



### HOSTED COATING PILOT



### 96 × 66 m / LUNAR SITE PLAN

Renew surfaces close to their use and build a practical record of lunar operation.

No crew on site / robotic operation  
Cargo: 1 hosted test delivery

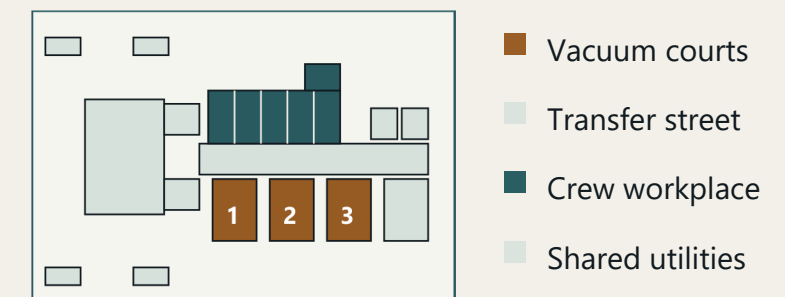
## 600 skins

Equivalent 50 m<sup>2</sup> surfaces  
Including repeated renewal

## 03 / SHARED WORKSHOP



### 1 Clean / 2 Prepare / 3 Coat



### 96 × 66 m / LUNAR SITE PLAN

Bring robotic production and skilled human repair into one shared workplace.

2 visiting specialists / 3 courts + 5 crew modules  
Cargo: 6–7 mass-only loads\*

## 17.3 t

Robotic equipment mass  
Equipment-only mass balance

13 / RESOURCES, PARTNERSHIPS AND PERMANENT-PRESENCE AMBITION

# An Earth Lab to launch a culture of repair.

## FIRST EXPERIMENT / FILM SAMPLES + PACKAGING

€38.6k planning budget before tax. A €27.1–62.4k range reflects workload and allowances. Public service rates form the basis.

| WORK PACKAGE           | RESOURCE BASIS                                 | BASE / EUR |
|------------------------|------------------------------------------------|------------|
| Coating + cleanroom    | 60 h × (€90 + €108) [B1, B2]                   | 11,880     |
| Technical support      | 60 h × €89 [B2]                                | 5,340      |
| Microscopy + optics    | 12 h × €60 + €2,000 allowance [B2]             | 2,720      |
| Thermal vacuum + packs | 24 h × €110; 18 h × €50; 6 × €40 [B3, B4]      | 3,780      |
| Protocol + review      | 24 h × €103.82 [B2]                            | 2,492      |
| Materials + transfers  | Project allowances for materials and logistics | 3,500      |

**€29.7k of work + 30% planning reserve = €38.6k. A potential €7k prize could launch the Lab, with €31.6k plus tax sought from partners.**

Months 0–2: partners, access and protocol. Months 2–5: film-sample trials. Next: a moving film ≥250 mm wide and a 5 × 10 m reflector demonstrator, with a dedicated budget.



## TOWARDS A PERMANENT PRESENCE

135 m<sup>2</sup> of shared crew workspace

Begin with two visiting repair specialists. Grow a shared maintenance service for reflectors and compatible settlement equipment. Resident teams would pair this workplace with sleeping quarters, continuous life support and resupply from the settlement.

Design for more than one life: demountable rails, serviceable cartridges, replaceable film and repairable frames. People carry the knowledge that keeps this material cycle working.

Planning budget, proposed partnerships and experiments / Design for disassembly: recover film, frames, cartridges, rails and power hardware for reuse or managed return.

# Sources, calculations and visual authorship

## 01 / SITE

NASA PGDA, south-pole Site 01 terrain products.

<https://pgda.gsfc.nasa.gov/products/78>

## 03 / CREW ALLOWANCES

H. W. Jones, Design and Analysis of a Flexible, Reliable Deep Space Life Support System (ICES 2012).

<https://ntrs.nasa.gov/citations/20160005782>

## 05 / LUNAR CARGO

NASA, Architecture Concept Review 2024: Lunar Surface Cargo.

<https://www.nasa.gov/wp-content/uploads/2024/06/acr24-lunar-surface-cargo.pdf>

## 07 / SHIELD REFERENCE

NASA ARMOR concept; Reprise support calculation summary on p. 6.

<https://ntrs.nasa.gov/citations/20130009378>

## 09 / HUMAN FIGURES

Renderpeople licensed scanned figures; commercial visualization use.

<https://renderpeople.com/free-3d-people/>

GLOBAL MOON / NASA Scientific Visualization Studio, LRO / Ernie Wright: <https://svs.gsfc.nasa.gov/4720/>

Architecture and machinery modelled to scale in Blender/Cycles. The hero is a native 3840 × 2160 render, with original suited figures; no generative image or upscaling is used in the submitted images. Technical studies are identified by a STUDY key. Terrain, licensed interior figures and materials are credited above. Calculations are preliminary; the proposed physical tests establish performance. The complete project is published at [marc-dvci.github.io/reprise](https://marc-dvci.github.io/reprise).

## 02 / HUMAN SYSTEMS

NASA-STD-3001 Vol. 2, natural and induced environments.

<https://www.nasa.gov/reference/6-0-natural-and-induced-environments-vol-2/>

## 04 / HABITABILITY

NASA Human Integration Design Handbook, revision 1.

<https://www.nasa.gov/human-integration-design-handbook/>

## 06 / VACUUM REFERENCE

V. Baglin and R. Kersevan, Vacuum systems, JUAS / CERN (2024). Reference equations, not Reprise measurements.

<https://cds.cern.ch/record/2929324>

## 08 / SURFACE MATERIALS

Poly Haven: Moon 03 and Moon Macro 01, CC0 scans of lunar simulant, not lunar samples.

<https://blog.polyhaven.com/moon/>

## 10 / VALUE + EARTH LAB PRICES

Jones / Take Material to Space or Make It There? (2023)

B1 / UPV NTC · PVD €90/h

B2 / URV · 2026 EXT tariffs

B3 / BIRA-IASB · TVAC €110/h

B4 / Packlab · vibration €50/h · 10 drops €40